AMINE RHAMMI

MECHANICAL ENGINEERING GRADUATE CANDIDATE | RELIABILITY & MAINTENANCE AND ASSET INTEGRITY

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UK Youth Mobility Scheme eligible | No sponsorship required

PROFESSIONAL PROFILE

B.Eng. Mechanical Engineering candidate graduating April 2027 combining 3 years of industrial experience through six placements across oil and gas, mining, pulp and paper, metallurgy, and manufacturing. Currently supporting mobile-equipment reliability at ExxonMobil Imperial Oil through maintenance readiness by combining SAP/CMMS evidence, RCM/FMECA/RCA, Weibull life-data analysis, condition monitoring, equipment strategy and practical field documentation. Earlier refinery and plant work covers pumps, valves, BOMs, MOC, P&IDs, commissioning, lubrication and CBM. Fluent in English/French and committed to progressing toward IMechE Chartership and a long-term UK engineering career.

EDUCATION

UNIVERSITÉ DU QUÉBEC À TROIS-RIVIÈRES (UQTR) — **Expected April 2027**

Trois-Rivières, Quebec, Canada

Bachelor of Engineering, Mechanical Engineering — Mechatronics concentration

CEAB-accredited engineering programme | Washington Accord recognised

Industry-Sponsored Capstone Project | Alcoa Corp — **Sep 2026 – Apr 2027**

Integrated Wire-Feed Handling and Control System for Aluminium Refining

Designing the measurement and control sub-system for an industrial wire-feeding application in molten-metal service, including sensor selection for real-time wire displacement, PLC-based drive control (GRAFCET/LADDER), slip-compensation logic, and alarm/interlock design for rupture, obstruction, and end-of-coil conditions. Responsible for calibration methodology, repeatability validation on a functional prototype, and integration of mechanical/electrical/logical interfaces

ROSEMONT COLLEGE — **Completed**

Montreal, Quebec, Canada

Diploma of College Studies (DEC), Mathematics & Computer Science

PROFESSIONAL ENGINEERING EXPERIENCE

IMPERIAL OIL | *EXXONMOBIL* |*Calgary, Alberta, Canada* — **May 2026 – Present**

Reliability & Maintenance Engineering Associate| 12-month student placement

- Led the data-quality workstream of a Kaizen project, consolidating more than 500 inconsistent symptom codes into 95 standardised, ISO 14224-aligned categories and improving coding accuracy from 70% to 80% against an 85% target.
- Designed and executed Reliability-Centered Maintenance (RCM) analysis on a 36-unit D11 dozer fleet, identifying undercarriage as the dominant cost driver escalating from 10% to 67% of fleet repair cost over 4 years and quantifying a 4.6× bad-actor spread across units to prioritize targeted intervention strategies.
- Developed an evidence-based OEM support assessment for CAT 797F structural tube assemblies, integrating inspection findings, installation histories, and product-support criteria across 73 haul trucks to classify 183 qualifying events and estimate CAD 1.4M in potential manufacturer-supported repair savings.
- Developed a Weibull-based maintenance-cost model for NRS coolers, quantifying a potential 26.9% cost reduction per avoided corrective replacement and evaluating the economic implications of a preliminary 4,000-hour intervention interval.
- Built a fleet-stewardship platform covering more than 40 mine-construction units across eight equipment types, decomposing equipment utilisation into physical availability, workforce availability and operational-efficiency losses while integrating maintenance cost and labour intensity to support model-level performance reviews.

VALERO Energy Corp. | *Montreal, Quebec, Canada* — **May 2025 – Aug 2025**

Mechanical Engineering Intern, Reliability | Fixed-term student placement

- Developed a brownfield hydrocarbon-drainage modification incorporating independent drainage routes, isolation valves and controlled air admission to maintain a jet-fuel tank within its vacuum limit during safe emptying.
- Prepared Management of Change (MOC) documentation, updated P&IDs, and authored commissioning checklists for a hazardous-service submersible-pump replacement, coordinated mechanical, structural, and electrical interfaces with the installation contractor to support safe start-up.
- Developed a low-pressure-drop magnetic pre-filtration concept for a hazardous-vapour system, targeting over 95% capture of ferrous particles above 10 µm at less than 1 inWC additional pressure drop.
- Reconciled field equipment information and OEM documentation against SAP Plant maintenance bills of material across pump families, identifying obsolete components, interchangeability opportunities, and master-data gaps affecting maintenance and spare-parts readiness.
- Performed field walkdowns across terminal loading stations to verify MOV and actuator nameplate data, building a terminal wide register that resolved tag gaps and supported risk-based spare-parts planning.

KRUGER INC. | *Trois-Rivières, Quebec, Canada* **May 2024 – Aug 2024**

Mechanical Engineering Intern, Reliability | Fixed-term student placement

- Supported turnaround execution for critical process equipment by reviewing inspection findings, converting emergent defects into repair scopes and coordinating technical actions through return-to-service verification.
- Prepared and helped conduct weekly condition-monitoring reviews identifying critical equipment, tracking repair actions and communicating lubrication requirements to technicians.
- Audited lubrication and preventive-maintenance routes for gaps coverage, prioritised critical equipment, and developed a standardised Excel tool to track lubrication points, lubricant requirements, task frequencies, and completion status.
- Standardised more than 9,000 roller-history records to calculate operating duration and developed a belt-management tool incorporating tension calculations for plant fans and agitators.

RIO TINTO | *Sorel-Tracy, Quebec, Canada* **May 2023 – Aug 2023**

Mechanical Engineering Intern, Reliability | Fixed-term student placement

- Designed mechanical fixtures, protective guards and process-piping modifications in SolidWorks and AutoCAD; completed bending-stress, deflection and weld-sizing calculations; and surveyed and field-verified elbow modifications designed to reduce pressure drop.
- Collected field evidence following a fire incident involving a water leak and supported implementation of a mitigation measure to reduce recurrence risk.
- Developed new preventive-maintenance plans for 14 conveyors using criticality assessments and OEM documentation, entering structured tasks and parts information into AssetWise and SAP PM.

SOUCY | *Drummondville, Quebec, Canada* **May 2022 – Aug 2022**

Mechanical Engineering Intern, Maintenance | Fixed-term student placement

- Coordinated annual overhead-crane and lifting-equipment inspections, defined asset scope, accompanied the external inspector and tracked identified deficiencies through corrective-maintenance closure.
- Developed an oil-waste handling and storage system with the site's environmental services provider, eliminating unsafe outdoor accumulation of open containers and reducing annual handling costs by more than CAD$ 30,000.
- Conducted troubleshooting on rubber-track production equipment faults, coordinating replacement-part requirements with suppliers and purchasing to support timely maintenance execution.

RIO TINTO | *Sorel-Tracy, Quebec, Canada* **May 2021 – Apr 2022**

Mechanical Engineering Intern, Process Optimisation| Fixed-term student placement

- Developed equipment-monitoring interfaces and supported root cause analysis (RCA), maintenance-plan development and compliance reviews for an upgraded-slag plant.
- Contributed to FMECA for the acid-regeneration plant, improving visibility of equipment failure modes, effects, and operational criticality.

CORE CAPABILITIES

Asset & operations optimization: Operational performance analysis; asset surveillance; lifecycle-cost analysis; equipment availability; performance improvement; risk-based prioritization; engineering decision support.

Facilities & mechanical engineering: Process facilities; pumps; compressors; heat exchangers; control valves/MOVs; piping; P&IDs; commissioning; field verification; equipment troubleshooting; ISO geometrical dimensioning and tolerancing.

Operations technology & digital engineering: PLC/GRAFCET; process automation; control systems; instrumentation; Python; SQL; Databricks; PI System/ProcessBook; Power BI; Excel/Power Query.

Capital projects & asset development: Project-delivery support; Management of Change (MOC); brownfield modifications; contractor coordination; economic evaluation; shutdown/turnaround planning; commissioning.

Reliability, risk & asset integrity: RCM; TapRooT RCA/RCFA; FMEA/FMECA; Weibull MLE and B10/B50; asset criticality; condition monitoring; PM/PdM; ISO 14224.

Engineering systems & field collaboration: SAP PM/CMMS; work orders and equipment histories; BOM/master data; spares planning; field walkdowns; multidisciplinary teamwork; technical communication.

SELECTED RELIABILITY & ENGINEERING TRAINING — EXXONMOBIL / IMPERIAL OIL

- TapRooT & Root Cause Analysis;
- Probability & Statistics for Reliability;
- RCM Equipment Strategy Development;
- Fixed Equipment Strategy Development;
- Risk-Based Work Selection Analysis;
- Pump, Compressor, Heat Exchanger and Control Valve Troubleshooting.